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Banking System Stress: Observable Conditions, Latent Vulnerabilities, and the First Pressure Points

Kilambi Research

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Abstract. This paper asks whether the global banking system is under observable stress as of 29 September 2026, and if not, where the first pressure points would appear. It reconstructs balance sheets for 56 banks across the United States, Europe, the United Kingdom, Canada, and Asia-Pacific from Q2 2026 and H1 2026 filings; measures the September 2026 macro-financial environment; audits asset quality and earnings buffers against the 2008–2009 benchmark; documents the 2023 digital-run failures from official autopsies; runs six hypothetical stress scenarios through the 26 banks with the most acute named vulnerabilities; maps three contagion paths and the post-2023 backstop landscape; and maintains a 13-indicator monitoring dashboard. The answer is a reported null: no observable system-level banking stress appears in current data. Capital sits above minima (US CET1 9.9 to 14.2%; Europe and Canada 12.9 to 17.2%), liquidity coverage exceeds 100% at every bank that discloses it, all 32 banks passed the June 2026 Federal Reserve stress test, the AT1 primary market is open, and buybacks are running at full pace. But the price environment is deteriorating: the US 10-year Treasury stood at 5.18% on 24 September, above its October 2023 peak; September brought a tightening wave (Federal Reserve 3.75–4.00%, ECB 2.50%, Bank of Japan 1.25%, RBA 4.60%); US bank equities fell 8 to 23% from summer highs; the MOVE index rose above 100; and weak-tail credit dispersion is the widest since 2022, while money-market gauges stay calm. The binding sequence in a shock is funding, not capital: a rate shock widens unrealized securities losses first (the drag on tangible common equity ranges from 4.6% to 40.6% in the US sample, with Bank of America the epicenter at $82.1bn), and uninsured-deposit flight then crystallizes the marks. No calibrated tail breaches the 4.5% CET1 minimum; the narrowest headroom is Societe Generale at 8.75%. The facility that separated liquidity stress from loss crystallization in 2023, the Bank Term Funding Program, closed in March 2024 with the last loan repaid in March 2025; the ECB's TLTRO matured in December 2024 with no successor; single-point-of-entry resolution remains untested doctrine. Three contagion paths are identified: US regional liquidity forcing HTM crystallization (the 2023 rhyme), the euro-area sovereign-bank nexus, and AI and datacenter credit repricing transmitted through private credit. Thirteen dashboard indicators show none at historical stressed levels, two at warning, and two within one move of warning. The paper states its falsifiers and its data gaps explicitly; the conclusion follows the data.

Keywords: banking; financial stability; bank runs; liquidity risk; stress testing; sovereign-bank nexus; commercial real estate; HTM securities; digital bank runs; systemic risk

JEL Classification: G21, G28, G32, G33, E44, E58

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1 Introduction

Is the global banking system under stress right now? The question matters because the price environment has deteriorated sharply into the autumn of 2026. The US 10-year Treasury closed at 5.18% on 24 September, above its October 2023 cycle peak; September delivered a coordinated tightening wave across the Federal Reserve, the ECB, the Bank of Japan, and the Reserve Bank of Australia; US bank equities fell 8 to 23% from summer highs; and bond-rate volatility, measured by the MOVE index, rose above 100. A natural inference from these prices is that bank balance sheets are under pressure. This paper tests that inference against current balance-sheet, funding, and credit data for 58 banks across five regions.

The answer is a reported null. Across the United States (16 banks), Europe and the United Kingdom (16), Canada (6), and Asia-Pacific (20), no observable system-level banking stress appears in current data (Observed, A). Capital ratios sit above regulatory minima, liquidity coverage exceeds 100% at every bank that discloses it, the June 2026 Federal Reserve stress test left all 32 tested banks above minimum capital requirements, the AT1 primary market is open with new issues in September 2026, and buybacks and dividends are running at full pace, which is management and supervisor revealed preference against current stress. The answer is not that nothing is wrong: the paper identifies six latent vulnerabilities, two warning-threshold dashboard breaches, and a deteriorating price environment. It is that the system looks capitalized and liquid today, and the paper reports that plainly rather than building a case.

The paper's spine is a four-bucket epistemic separation, applied to every finding. Bucket (a), observable stress: facts in current data, such as realized losses, funding outflows, and ratio breaches. Bucket (b), latent vulnerability: exposures that are disclosed but not yet loss-bearing, such as unrealized securities losses and uninsured-deposit concentrations. Bucket (c), hypothetical tail: model outputs under stated scenarios, including the Federal Reserve and EBA stress tests and the desk's six-scenario exercise, labeled as such and never presented as observation. Bucket (d), excluded speculation: claims the evidence cannot support, which are listed in the gaps sections and not made. Sections below are organized so the reader can always tell which bucket a claim sits in.

The contributions are as follows. First, a bank-by-bank reconstruction of 58 institutions from Q2 2026 and H1 2026 filings, with every metric graded by source quality and epistemic bucket (Section 4). Second, a September 2026 macro-financial environment measurement that separates stressed prices (long-end yields, bank equities, rate volatility, weak-tail credit) from calm plumbing (overnight funding, euro-area term funding, aggregate spreads) (Section 3). Third, an asset-quality and profitability audit against the 2008–2009 benchmark: pre-provision earnings cover net charge-offs 6.4 times, versus 1.33 times at the 2009 trough; bank-book CRE delinquency is 1.50%, versus 9.13% at the GFC peak; the CMBS market shows the stress that bank books have not yet recognized (Section 5). Fourth, verified 2023 failure autopsies (SVB, Signature, First Republic, Credit Suisse) used to calibrate what a digital run looks like in official records (Section 6). Fifth, six hypothetical stress scenarios run through the 26 banks with the most acute named vulnerabilities, establishing that the binding sequence is funding first, capital second, and that no calibrated tail breaches the 4.5% CET1 minimum (Section 8). Sixth, three contagion paths and a tested-versus-assumed inventory of backstops, including the finding that the Bank Term Funding Program, the instrument that separated liquidity stress from loss crystallization in 2023, no longer exists (Section 9). Seventh, a 13-indicator monitoring dashboard with stated warning thresholds (Section 10). The paper closes with falsifiers (Section 11), limitations and data gaps (Section 12), and the conclusion (Section 13).

This paper is the third in a sequence. KR-WP-2026-10, "AI Capex and the Corporate Credit

Market," (Kilambi Research, 2026a) analyzes how the AI infrastructure buildout is financed and what it implies for credit; KR-WP-2026-11, "Can the AI Buildout Pay for Itself?" (Kilambi Research, 2026b) models the demand side. Neither contains bank balance-sheet analysis; the AI and datacenter exposure of banks is a contagion path here (Section 9.3), cross-referenced to KR-WP-2026-10 rather than duplicated.

The rest of the paper proceeds as follows. Section 2 reviews the related literature. Section 3 measures the macro-financial environment. Section 4 presents the bank-by-bank reconstruction for the United States, Europe and Canada, and Asia-Pacific. Section 5 audits asset quality and profitability. Section 6 documents digital-run dynamics and the 2023 autopsies. Section 7 frames the current position against 2008–2009. Section 8 runs the six stress scenarios. Section 9 maps contagion paths and backstops. Section 10 presents the monitoring dashboard. Sections 11 and 12 state falsifiers and limitations. Section 13 concludes. A data appendix documents series definitions and reproduction files.

2 Related literature

The theory of runs starts with Diamond and Dybvig (1983): banks fund illiquid assets with demandable deposits, and a shift in depositor expectations can move the system from the good to the bad equilibrium. The 2023 failures are a near-laboratory realization of that mechanism at digital speed. Gorton and Metrick (2012) extended run theory to wholesale funding, documenting how the 2007–2009 crisis ran through repo collateral rather than retail deposits; the US regional episode of 2023 ran through uninsured corporate deposits and, crucially, through the realization of unrealized securities losses, which is the funding-capital interaction this paper's scenarios trace.

The empirical 2023 literature is now substantial. The Federal Reserve's own review of Silicon Valley Bank (Barr, 2023) called it "a textbook case of mismanagement" and found supervisors had not appreciated the bank's vulnerabilities as it grew. The FDIC's supervision report on Signature Bank (Federal Deposit Insurance Corporation, 2023) traced the failure to poor management and rapid unrestrained growth, with management rejecting examiner concerns about uninsured-deposit stability as late as noon EST on March 10, 2023. Cookson et al. (2023) document the social-media acceleration channel: withdrawals coordinated over messaging apps and venture-capital networks in hours, consistent with the FDIC staff finding that the vast majority of outflows left by wire and that daily outflow rates above 20% of deposits are unprecedented in its records. This paper uses these autopsies as calibration for what a run costs in time and dollars (Section 6), not as a forecast input.

On funding under tightening, Drechsler et al. (2017) establish the deposits channel of monetary policy: banks with market power reprice deposits slowly, widening net interest margins as rates rise. The desk's cost-of-funds beta reconstruction for 2026 (commercial-bank betas 0.45 to 0.76, intact NIMs; Section 3.4) sits squarely in this channel's logic: margins look healthy now precisely because deposit repricing lags the policy rate, which is also why the cost base is exposed if rates stay high.

The sovereign-bank nexus is covered by Acharya et al. (2014), who show how bank bailouts transferred risk to sovereigns and back to banks in the euro-area crisis, and by Brunnermeier et al. (2016) on the diabolic loop between sovereign and bank balance sheets. Section 9.2 traces the 2026 version: domestic sovereign concentrations of 3 to 7% of exposure at Italian and Spanish banks, held largely at amortised cost and performing, which is a concentration to monitor rather than current stress.

On stress testing, Acharya et al. (2018) survey the post-crisis regime and its limitations, including

scenario design and the absence of a predefined pass-fail threshold in the EBA exercise, both of which this paper discloses rather than works around (Sections 4.3, 8.3). The supervisory scenarios themselves, the Federal Reserve 2026 exercise (Board of Governors of the Federal Reserve System, 2026) and the EBA 2025 EU-wide test (European Banking Authority, 2025), are used here strictly as hypothetical tails: they tell us where capital would go under a severe macro path, not where it is going.

3 Macro-financial environment

Policy is restrictive nearly everywhere, and September 2026 tightened it further in three of the four largest advanced economies. The reference date for market readings is 29 September 2026 unless stated. Every market number below carries an evidence grade; the full series archive is listed in the data appendix.

3.1 Policy rates

The Federal Reserve raised the target range to 3.75–4.00% (IORB 3.90%) on 16 September (OBSERVED, A); the ECB raised the deposit facility rate to 2.50% on 10 September (MRO 2.65%) (OBSERVED, A); the Bank of England held at 3.75% on 17 September by a 6–3 vote (OBSERVED, B); the Bank of Japan raised the overnight call target to 1.25% on 18 September by 7–2, the highest since 1995 (OBSERVED, B); the RBA raised to 4.60% on 29 September, its fourth hike of 2026 and the highest since 2011 (OBSERVED, B). The PBoC held the one-year LPR at 3.00% for a sixteenth consecutive month and the Bank of Canada held at 2.25% for a seventh consecutive meeting (OBSERVED, B). Every advanced-economy rate shown is near the top of its post-2008 range (Figure 1); the two tightening outliers are the RBA, still hiking into September, and the BoJ, which normalized from negative territory to 1.25% within roughly two years.

3.2 Long-end yields

Long-end yields, not policy rates, are the most stressed part of the curve. The US 10-year Treasury stood at 5.18% on 24 September (OBSERVED, A), above the October 2023 cycle peak of 4.98% and the highest since the pre-crisis era; the 2-year stood at 4.87%, below its 2023 peak of 5.19%, so the curve disinverted to +31bp 2s10s. The euro-area AAA 10-year stood at 3.63% on 28 September, its 81st percentile since 2004 and above its 2022–2023 peak of 3.03% (OBSERVED, A), with 2s10s at +40bp (Figure 2). High long-end yields widen unrealized losses on bank securities books; that is the mechanical link between this section and Sections 4 and 8.

3.3 Bank equities and credit spreads

Bank equities are the one broad market price showing moderate stress. Measured against each name's 2026 high at the 29 September close (OBSERVED, A): Capital One −22.9%, Goldman Sachs −20.8%, Morgan Stanley −15.3%, Wells Fargo −15.0%, Bank of America −14.7%, Deutsche Bank −13.3%, the KBW index −11.5%, and JPMorgan −8.3%, against an S&P 500 drawdown of only −1.6%. European banks are less affected: UniCredit −2.4%, Intesa −2.0%, Santander −3.0%, HSBC −6.7% (Figure 3). The pullback is concentrated in US names and in wholesale and

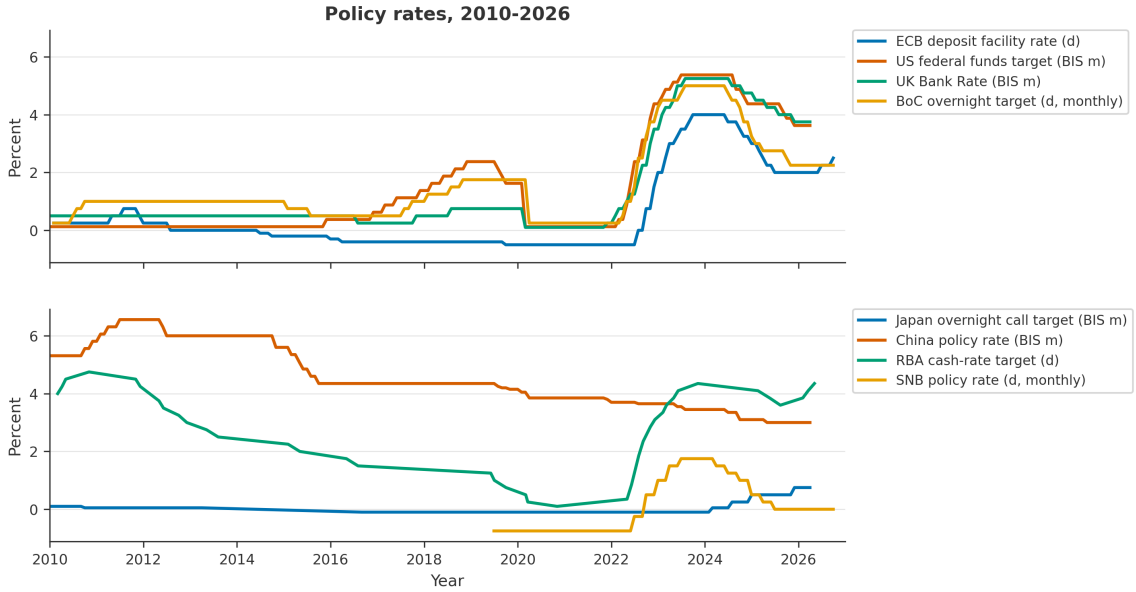


Figure 1. Policy rates, 2010–2026. The September 2026 tightening wave pushed the Fed to 3.75–4.00%, the ECB to 2.50%, the BoJ to 1.25% (highest since 1995), and the RBA to 4.60% (highest since 2011).

Source: BIS monthly policy-rate mirror through April 2026 and central-bank releases thereafter. Note: BoE, BoJ, PBoC, BoC, and RBA September readings are press reporting of official releases (Observed, B); Fed and ECB are primary sources (Observed, A).

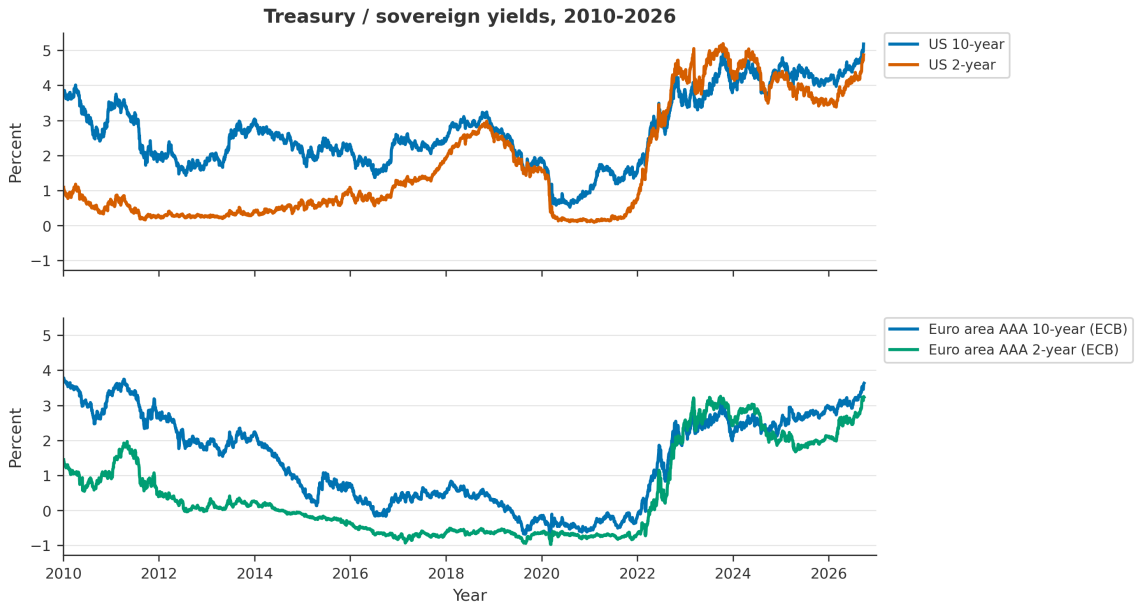


Figure 2. Benchmark sovereign yields. The US 10-year reached 5.18% on 24 September 2026, above its October 2023 peak; the euro-area AAA 10-year reached its 81st percentile since 2004.

Source: FRED H.15 mirror (US), ECB mirror (euro area). Note: series through September 2026; the 2s10s disinversion is derived from the same sources.

capital-markets-heavy business models, and it followed summer 2026 all-time highs rather than any funding event. That pattern is repricing, not distress pricing.

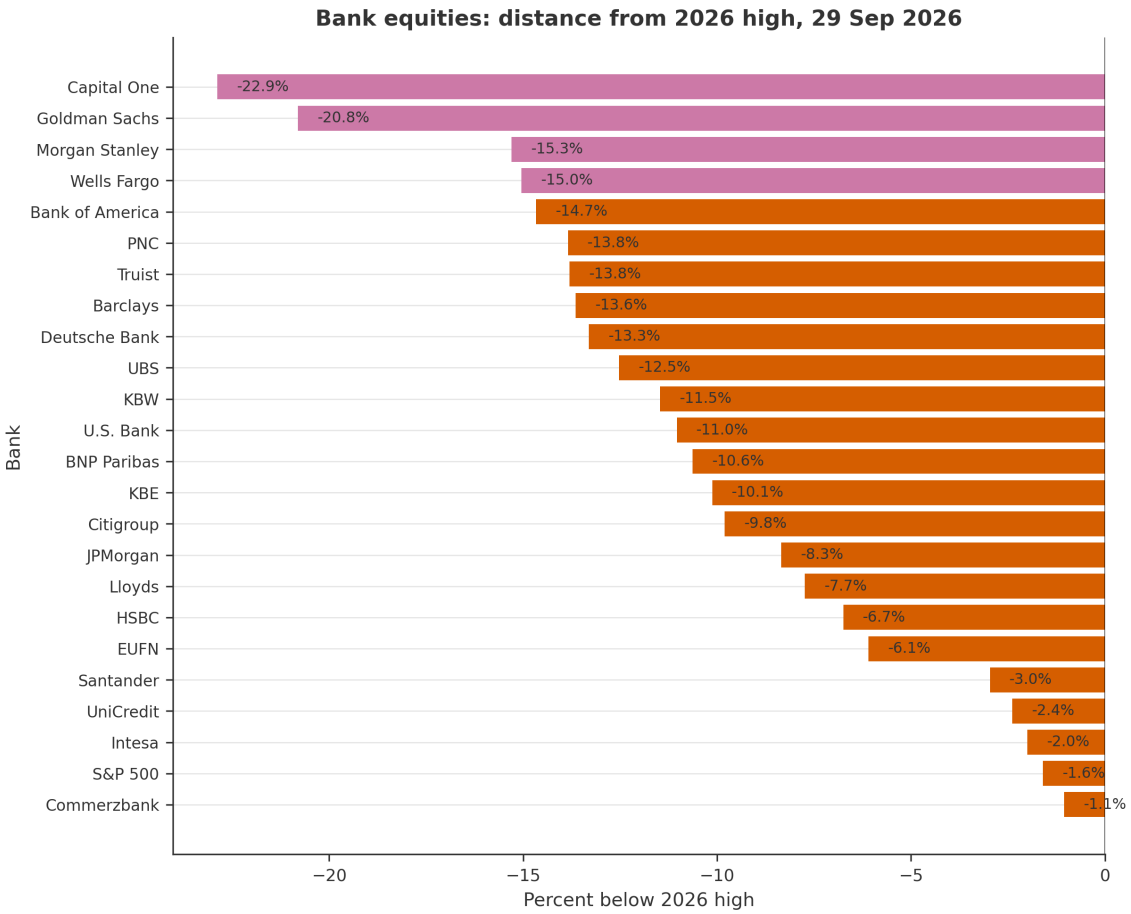


Figure 3. Bank equity drawdowns from 2026 highs, 29 September 2026 close. US names are down 8 to 23%; European names 2 to 7%; the S&P 500 is down 1.6%. The pullback followed summer all-time highs, not a funding event.

Source: Yahoo Finance, 29 September 2026 close (Observed, A).

Credit spreads are bifurcated. US investment-grade OAS was about 77bp and high-yield about 282bp on 25 September, near multi-year tights (Observed, B). But the CCC-minus-BB differential was about 787bp, its widest since the 2022 hiking cycle (Observed, B): aggregate spreads say calm, the weakest tail says stressed. Bond-rate volatility is elevated, with the MOVE index above 100 on 29 September after a roughly 29% weekly rise (Observed, B). For European bank credit, the only archived quote is iTraxx Europe Senior Financials at 54bp in January 2026 (Observed, B); no verified September quote was located, and single-name bank CDS are proprietary. That gap is stated in Section 12 and carried on the dashboard as a missing leg.

3.4 Funding plumbing: not stressed

Overnight funding shows no stress. SOFR was 3.90%, EFFR 3.88%, and OBFR 3.88% on 25 September (Observed, A); SOFR minus EFFR was +2bp and OBFR minus EFFR 0bp. The

SOFR tail gauge (99th percentile minus median) was 9bp, against 375bp in the September 2019 repo episode and 30bp at its 2022–2023 maximum. The 2019 and March 2020 spikes remain visible in the tail series, so the gauge is not structurally suppressed; it is simply quiet now. Euro-area bank term funding is also calm: three-month EURIBOR was 2.51% in August, 26bp above the ECB deposit facility rate (Observed, A), versus a 2022–2023 peak of 108bp. This is a normal level, not a stress reading.

Bank funding costs have repriced but remain below full pass-through at commercial banks. Cumulative deposit betas from 2022-Q1 to 2026-Q1, measured at the FDIC-certified bank level as annualized interest expense over deposits relative to the move in the effective fed funds rate, range from 0.45 (Wells Fargo) to 0.76 (Capital One); Goldman Sachs and Morgan Stanley exceed 1.0 (1.08 each), reflecting wholesale-funded business models (Reconstructed, B). These are cost-of-funds betas: the input is total interest expense, not deposit-only expense, since FDIC EINTEXP includes wholesale interest expense. NIM proxies are intact: 1.30–2.83% for commercial and universal banks, 7.28% for Capital One, 1.30–1.38% for the wholesale-funded dealers. In the terms of Drechsler et al. (2017), the deposits channel is doing what it predicts: margins look healthy because deposit repricing lags the policy rate, which leaves the cost base exposed if rates stay high. That exposure is a latent vulnerability, not current stress.

The pattern of this section is consistent: price stress in duration and in bank equity valuations, with funding plumbing functioning normally. That is the opposite of 2008 and 2020, when funding spreads led, and closer to 2022–2023, when rate moves led and funding followed only in the March 2023 episode. The excluded-speculation list for this section: no claim is made that AI investment has caused banking stress (KR-WP-2026-10 (Kilambi Research, 2026a) found credit aggregates expanding and spreads narrowing through the AI capex boom, and nothing in the banking data contradicts that), and recession timing, election outcomes, and future policy paths are out of scope.

4 Bank-by-bank reconstruction

The desk reconstructed 56 banks from Q2 2026 and H1 2026 filings, SEC XBRL companyfacts, FDIC SDI lead-bank data, and the June 2026 Federal Reserve stress-test results: 16 US banking organizations, 16 European and UK banks, 6 Canadian banks, and 18 Asia-Pacific banks. Every metric carries an evidence grade: A for primary filings, B for XBRL or reputable secondary, C for desk arithmetic from primary values, D for FDIC lead-bank subsidiary proxies or unverified press. Cutoff is filings and disclosures through Q2 2026; Canadian banks report Q3 FY2026 ended 31 July 2026. Table 1 summarizes the sample. The headline finding across all four regions is the same: capital above minima, liquidity coverage above 100% where disclosed, and no observable system-level stress.

4.1 United States

The null holds for the 16-bank US sample. The June 2026 Federal Reserve stress test, released June 24 and applying a severe scenario with deep CRE, housing, and corporate-credit losses over 2026Q1 to 2028Q1, left all 32 tested banks above minimum capital requirements (Observed, A) (Board of Governors of the Federal Reserve System, 2026); aggregate CET1 fell from 12.8% to a projected minimum of 11.2%. Scenario-minimum CET1 for the sample runs from 8.6% (Citizens) to 12.6%

Table 1. Bank sample: coverage and headline ratios, Q2/H1 2026

Region	Banks	CET1 ratio	LCR (disclosed)	Asset-quality anchor
United States	16	9.9–14.2%	107–165% (7 of 16)	CRE delinquency 1.50% (Q2 2026)
Europe / UK	16	12.9–17.2%	125–161%	EBA NPL 1.8% (Q4 2025)
Canada	6	13.0–13.6%[a]	125–141%	90+ day mortgage arrears 34–51bp
Asia-Pacific	18	9.4–23.8%	113–142%[b]	NPL 0.8–1.6% (bank-level)

Notes: CET1 ranges are reported ratios from Q2/H1 2026 filings (Observed, A/B). The US CET1 median is 10.8% (Reconstructed, B). (a) Canadian CET1 range from bank Q3 FY2026 disclosures. (b) Observed only for DBS (142%), CCB (138%), and disclosed NSFR for OCBC/CCB; bank-level LCR is not retrieved for most APAC banks, a stated gap. Source: desk reconstruction from bank filings, FDIC SDI, ECB/EBA aggregates; see data appendix.

(JPMorgan). These are hypothetical model outputs; they support the null but do not constitute observed resilience.

Securities valuation drag is the largest latent vulnerability. Combined AOCI and unrecognized HTM losses as a share of tangible common equity range from 4.6% at M&T to 40.6% at Bank of America, which carries an $82.1bn unrecognized HTM loss, the largest in the sample (Observed, A) (Figure 4). For Category I banks, AOCI already flows into CET1, so the regulatory number reflects it; the HTM slice sits outside capital everywhere. These are unrealized marks on hold-to-maturity portfolios, not credit losses, and they shrink as the bonds roll off. The risk they represent is liquidity-event risk: a forced sale crystallizes them.

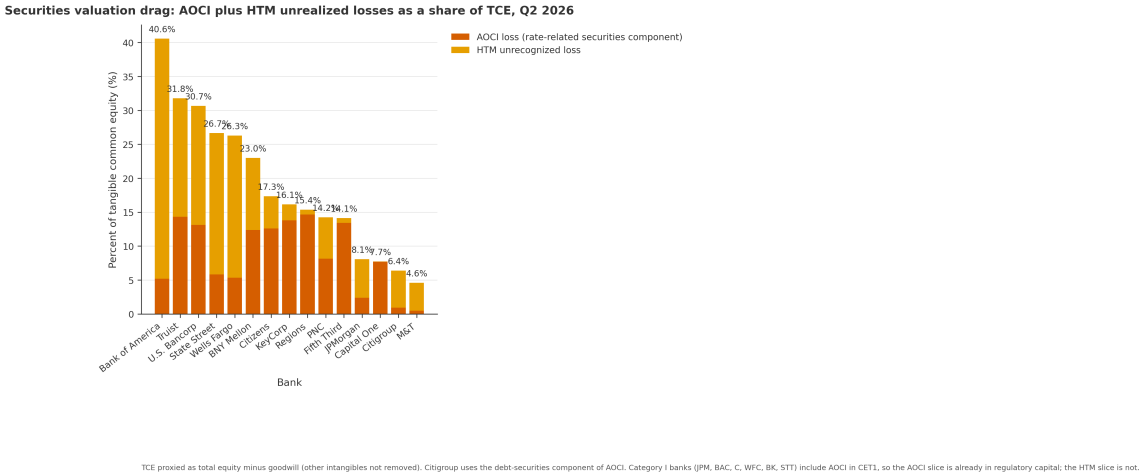


Figure 4. Securities valuation drag: unrecognized HTM losses plus AOCI as a share of tangible common equity, Q2 2026. Bank of America is the epicenter at 40.6% ($82.1bn HTM loss); M&T is lowest at 4.6%. These are unrealized marks, not credit losses; they crystallize only under forced sale.

Source: Q2 2026 10-Q filings (Observed, A); TCE computed by the desk from filing values (Reconstructed, C). Note: for Category I banks AOCI already flows into CET1; the HTM slice sits outside capital everywhere.

Uninsured deposits versus liquidity buffers show no breach. Eight banks disclose both an uninsured share and an LCR; every one sits above the 100% LCR minimum, and Capital One pairs the lowest uninsured share (15%) with the highest LCR (165%) (Observed, A) (Figure 5). Two measurement cautions: Citi's 82.5% uninsured share is an FDIC lead-bank proxy (Reconstructed, D), and

State Street's FDIC ratio exceeds 100% on a denominator artifact, so no consolidated figure is plotted for it; the company publishes no consolidated ratio. The custodians (BNY Mellon 99.3%, State Street) run the highest uninsured shares by construction; their deposits are institutional operating balances, and both hold LCRs above 105%.

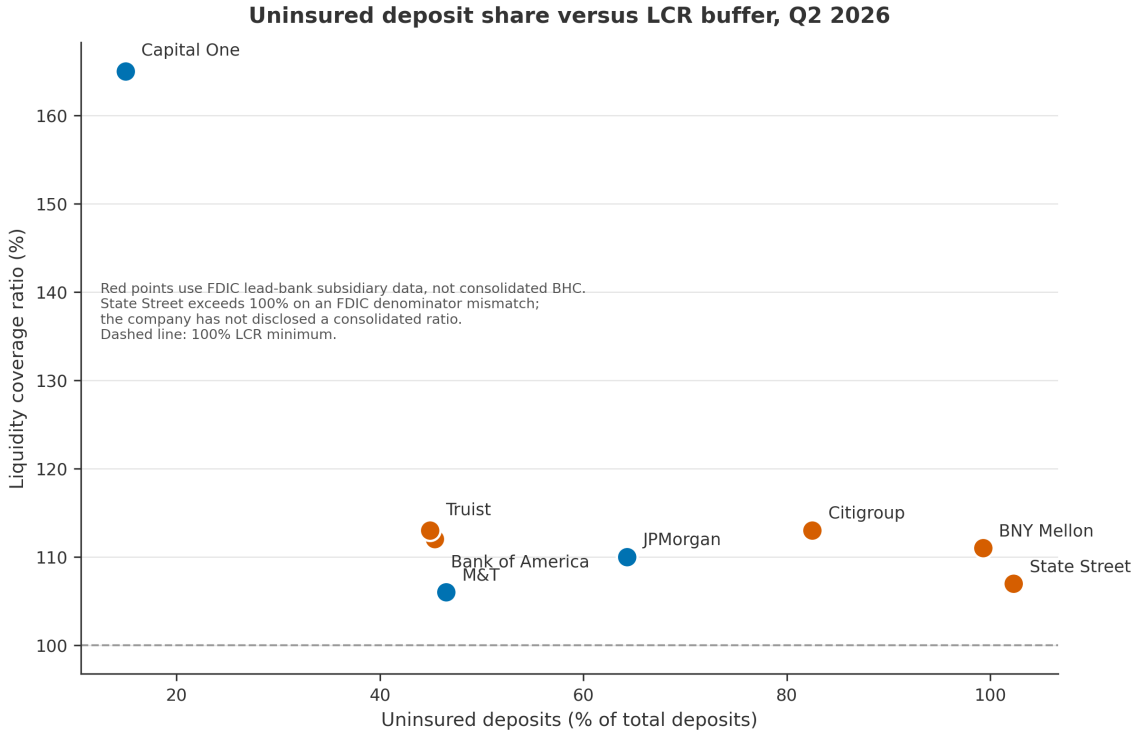


Figure 5. Uninsured-deposit share against LCR, Q2 2026. All banks disclosing both sit above the 100% LCR minimum. Citi's 82.5% is an FDIC lead-bank proxy, not a company disclosure.

Source: Q2 2026 10-Q filings and FDIC SDI. Note: uninsured shares are company-stated (A), computed from filings (C), or FDIC lead-bank proxies (D); the custody banks' shares are structural to their business models.

Office CRE is the one pocket where realized deterioration is visible, and it is small, reserved, and shrinking. Six of sixteen banks disclose an office figure, from $858m at Regions to $14.7bn at JPMorgan; absence of disclosure is not evidence of zero exposure. Bank of America has $2.8bn criticized office on $11.4bn of office loans (24.6%), down 19% year over year with $58m of H1 office net charge-offs (Observed, A); Regions has $121m nonperforming on $858m (14.1%), but office is 0.9% of total loans and H1 charge-offs were $1m.

Narrower current CET1 headroom, worth naming rather than alarming over: PNC (9.9%), Fifth Third (9.93%), Huntington (10.0%), Wells Fargo (10.26%), and M&T (10.19%) sit below peers above 11%, though all clear their stress-test minima with margin. Wholesale and FHLB reliance concentrates at Truist ($17.6bn FHLB advances), Citizens ($11.6bn utilized of $30.5bn capacity), M&T ($4.6bn), and Huntington ($27.6bn wholesale funding, up $3.2bn). Gaps not estimated: portfolio duration and hedge coverage, wholesale maturity schedules through 2028, and office exposure for the ten non-disclosing banks.

4.2 Europe, United Kingdom, and Canada

No observable banking stress in this 22-bank set as of 29 September 2026 (Observed, A/B). Reported CET1 ratios run 12.90% (BBVA) to 17.20% (Credit Agricole Group), and every bank with a disclosed requirement clears it with margin: BBVA 12.90% against an 8.98% SREP requirement, Intesa Sanpaolo 13.1% against 9.96%, Nordea 15.4% against 13.8%. The EBA Q4 2025 aggregate CET1 is 16.3%.

The binding revealed-preference test is capital distribution, and it is running at full pace: HSBC reinstated a $1bn buyback; Barclays £1bn; Lloyds £1bn; BBVA completed €4bn and launched a further €2bn; Intesa guides €9.4bn total 2026 shareholder return; Societe Generale an extraordinary €1.5bn buyback from August 2026; Deutsche Bank a new €500m buyback; TD plans C$13bn of FY2027 buybacks. Supervisors would not permit this pace if they saw current stress. This is evidence against observable stress, not for it. Liquidity reads the same: LCRs of 125% (RBC) to 161% (Barclays), NSFR 122% to 138% where disclosed, EBA aggregates of 163.1% and 126.9%, and TLTRO III fully matured on 18 December 2024 with 2026 balances at zero, so current positions are post-TLTRO and self-funded.

Asset quality is normalizing, not stressed. The EBA aggregate NPL ratio is 1.8% (Q4 2025) with Stage 2 loans at 9.1%; bank cost of risk runs 17bp (UniCredit) to 56bp (Scotiabank). Canadian 90+ day mortgage arrears are rising but low in level: CIBC 51bp (highest of the six), RBC 41bp, Scotiabank 34bp. CIBC shows the clearest mortgage deterioration, with 0.51% total 90+ delinquency versus 0.36% a year earlier and C$1.366bn net impaired mortgages (Observed, A): credit normalization, not a capital event. Profitability is strong, with RoTE/ROE mostly 11–25%, and the junior-subordinated primary market is open: Deutsche Bank announced a benchmark AT1 on 21 September 2026 and UniCredit issued €750m at 6.25% to December 2031 (284.1bp reset margin) on 18 September (Observed, B). A pulled or distressed AT1 issue would be a market-based stress signal; the market is printing.

The latent vulnerabilities are three. First, the Canadian mortgage renewal and repricing wave: TD has 46% variable-rate residential real-estate lending, the highest variable share of the six, and CIBC illustrates the arithmetic, with FY26Q4 renewals facing +$243/month (+11%) at an illustrative 4.5% renewal rate and C$88bn of mortgages renewing over FY26Q4–FY27Q4 (Reconstructed, B). The vulnerability is household cash flow and credit migration, not bank solvency at current levels; it becomes a bank problem only if unemployment rises materially alongside. Second, domestic sovereign concentration, the sovereign-bank nexus of Acharya et al. (2014) and Brunnermeier et al. (2016) in 2026 form (Figure 6): BBVA 6.84%, UniCredit 5.21%, Intesa Sanpaolo 4.85%, Societe Generale 4.25%, Credit Agricole 3.80%, Santander 3.28%, BNP 1.94% of exposure, a lower bound computed against December 2024 exposure with a vintage-mismatched denominator. Holdings are largely amortised-cost and performing: a concentration to monitor, not current stress. Third, unrealized securities losses: the last supervisory aggregate is the ECB/SSM ad-hoc report of 28 July 2023, €73bn net unrealized on amortised-cost books at February 2023 (gross €116bn, €40bn offset by hedges; about 60% of yield-curve risk hedged, spread risk mostly unhedged). No 2026 supervisory aggregate was found, so any claim about 2026 EU unrealized losses is labeled inference, not observation.

The EBA 2025 EU-wide stress test (published 1 August 2025) (European Banking Authority, 2025) is a hypothetical tail, not a forecast: an adverse scenario to end-2027 depletes aggregate CET1 by 370bp to 12.0% with €547bn cumulative losses. Bank-level adverse CET1: Societe Generale 8.75% (454bp depletion, the narrowest in the set), BNP 9.49%, Deutsche Bank 10.23%, ING 10.63%,

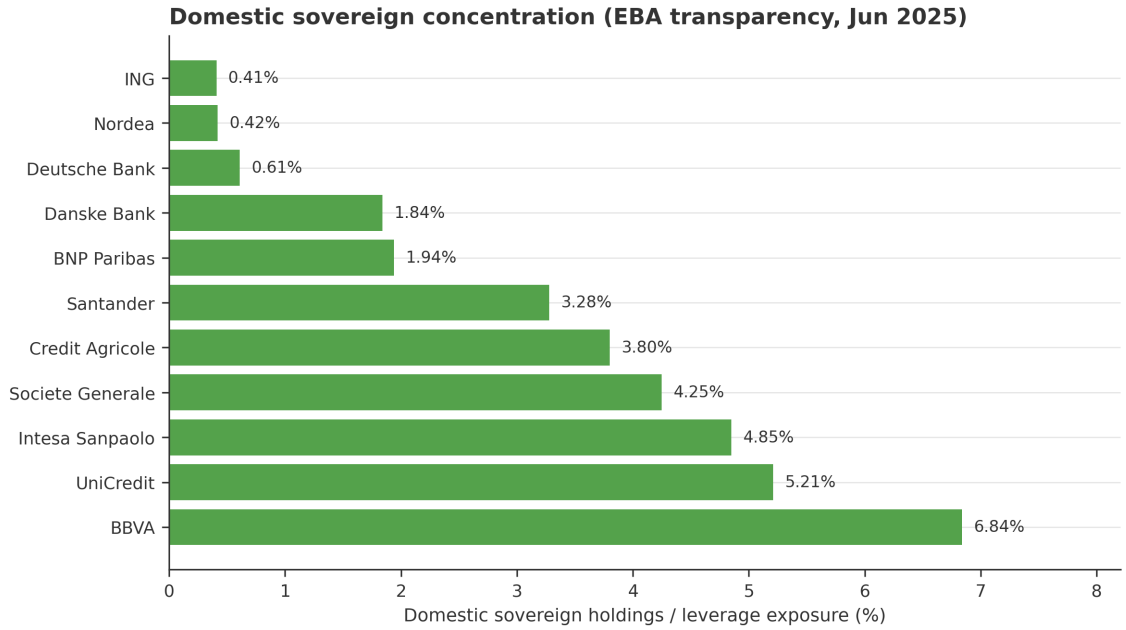


Figure 6. Domestic sovereign holdings as a share of exposure, June 2025. The bank-sovereign nexus is most visible in Italy and Spain; holdings are largely amortised-cost and performing.

Source: EBA transparency exercise, June 2025 holdings over December 2024 exposure (a lower bound; vintage-mismatched denominator) (Observed, B). Note: ratio uses exposure, which exceeds total assets.

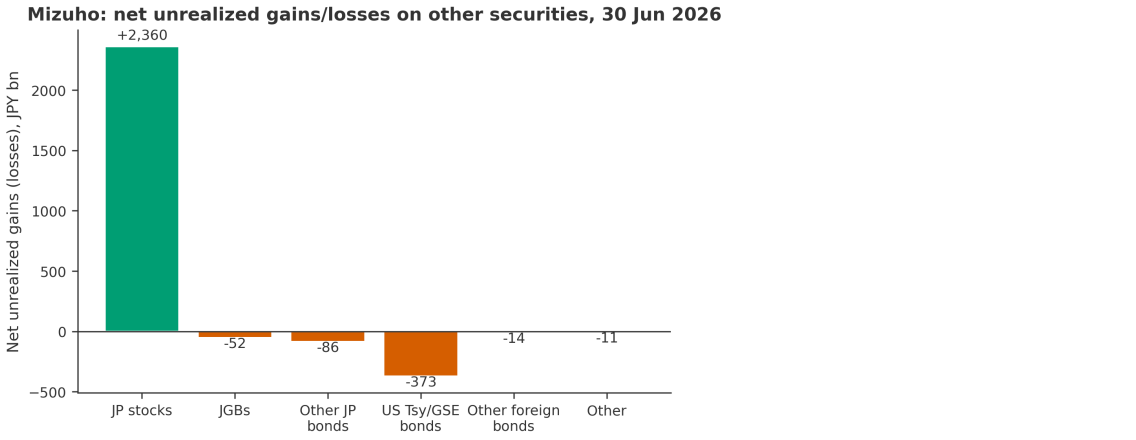
Santander 10.91%, Credit Agricole 11.01% (617bp depletion, the largest), BBVA 11.02%, Intesa 12.00%, UniCredit 12.50%, Nordea 12.62%, Danske 13.56%. Two cautions: the test has no predefined pass-fail threshold, so the claim that all banks "stay above requirements" was not demonstrated against bank-specific MDA thresholds and is not made here; and adverse-scenario profit turns negative in 2025 for most banks before recovering by 2027. UK banks (HSBC, Barclays, NatWest, Lloyds) and UBS are not in the EBA test, so no comparable adverse numbers exist for them.

4.3 Asia-Pacific

No bank in the APAC sample shows combined solvency, liquidity, and funding stress as of the latest disclosures, mostly H1 2026 (Observed, A/B). Observable pockets of stress are narrow; the largest risks are latent, chiefly data opacity around Chinese property and LGFV exposures.

Japan's megabanks carry observable unrealized foreign-bond losses that do not currently stress solvency. Mizuho's AT1 prospectus gives the cleanest decomposition at 30 June 2026: foreign-bond unrealized losses of −387.0bn JPY (of which −372.9bn US Treasury and GSE bonds) and Japanese-bond losses of −138.8bn (JGBs −52.5bn), against Japanese-stock gains of +2,359.6bn JPY, for a net position of +1,822.5bn JPY after deferred hedges (Observed, A) (Figure 7). The losses are real and foreign-bond driven, but small relative to equity and offset by the stock book. MUFG (CET1 12.95% at June 2026) and SMFG (12.63%) disclosed no equivalent June 2026 breakdown in the sources retrieved. MUFG's CET1 buffer above requirement narrowed to about 50bp at March 2026, but the driver was the Shriram Finance acquisition, not bond losses. Megabank earnings

remain strong, with net interest income up 18.6% in FY2025 on the BoJ rate cycle. A sector-wide FSA figure of 7.05tn JPY unrealized JGB losses is cited in press but could not be verified from a primary FSA source; it is retained at grade D and excluded from the analysis.



Source: Mizuho AT1 prospectus (424b5, stocktitan). Net total +1,822.5bn incl. deferred hedge. MUFG/SMFG did not disclose an equivalent JGB breakdown in sources found.

Figure 7. Mizuho unrealized securities position, 30 June 2026. Foreign-bond losses of −387bn JPY are real but small relative to equity and are offset by the Japanese-stock book (+2,360bn JPY). Net position after hedges: +1,823bn JPY.

Source: Mizuho AT1 prospectus, 30 June 2026 (Observed, A). Note: MUFG and SMFG disclosed no equivalent June 2026 breakdown in retrieved sources; the FSA sector-wide figure is unverified (grade D) and excluded.

China's observed fact is margin compression; its caveat is credit opacity. ICBC's net interest margin fell from 2.11% in 2021 to 1.28% in 2025 and was 1.29% in H1 2026, an 83bp collapse in five years with no visible bottom (Observed, A); CCB, ABC, and BOC all report H1 2026 NIMs of 1.27 to 1.37%. Against this, headline asset-quality metrics are stable: ICBC NPL 1.31% with 213.6% coverage, CCB 1.33% with 239.4%, ABC 1.25% with 290.1%. The divergence is property-specific: CMB's corporate real-estate NPL ratio was 4.47% (versus 1.16% retail) on CNY14.6bn, CITIC Bank cut property exposure 7% in H1 2026 to CNY353bn, and CMB's retail NPL ratio rose quarter on quarter. Capital remains ample by reported ratios (ICBC CET1 13.89%, CCB 14.34%, ABC CAR 17.50%), and ABC announced a CNY160bn A-share placement to replenish CET1. The disclosure-quality caveat applies to every China row: no sample bank discloses a comparable LGFV exposure line, so reported figures are observed quantities on a potentially incomplete loss surface, and no Chinese estimate is presented as observed data beyond what filings state.

Hong Kong commercial real estate is the one pocket of realized credit stress in APAC. Hang Seng reported HK$25bn of impaired Hong Kong CRE loans at H1 2025, roughly half of HSBC's Hong Kong CRE book; HSBC booked a further US$0.2bn Hong Kong CRE expected-credit-loss charge in H1 2026, down from US$0.5bn in H1 2025. The $3bn portfolio-sale process indicates active balance-sheet repair rather than funding pressure. BOCHK's 0.89% impaired-loan ratio and 23.80% CET1 show no solvency stress. In Singapore, UOB is the outlier (NPL 1.6% versus 1.0% at DBS and 0.9% at OCBC, S$615m of general CRE provisions, over 40% of its Hong Kong branch loans property-related); DBS (CET1 16.6%, ROE 17.5%) and OCBC (CET1 15.7% transitional, 14.0% fully phased) posted record Q2 profits. In Australia, APRA aggregates show riskier origination and rising arrears, not a bank-capital event: interest-only loans reached 23.5% of new housing lending in Q2 2026 (highest since the series began in 2019), serviceability-policy exceptions hit a record 5.8%,

and mortgages 90 or more days past due reached A$25.9bn or 1.01% of mortgages, above the 0.93% average since 2019 (Observed, A). Bank CET1 ratios are steady at 11.6 to 12.4%, and APRA's February 2026 debt-to-income cap is the binding backstop. Classification: latent vulnerability trending the wrong way, with observable arrears movement but no observable bank stress.

5 Asset quality and profitability

Realized-loss absorption capacity is strong at the aggregate level; bank-book delinquencies are far below GFC peaks; CRE stress is concentrated in office and in CMBS rather than in bank delinquency rates so far; the 2026–2027 maturity wall is large and bank-held; lending standards have normalized to neutral; and bank linkages to non-bank financial intermediaries keep growing. What this section does not show: the distribution of stress across banks, latent losses not yet charged off, or anything about private-credit warehouse and subscription-line exposures, for which no official aggregate exists and which must not be inferred from its absence.

5.1 Earnings buffers

US aggregate pre-provision net revenue covered net charge-offs 6.4 times in H1 2026 and 5.9 times in 2025 (Observed, A), in line with the post-crisis norm of 5–7 times since 2014 and far above the GFC trough of 1.33 times in 2009 (Figure 8). This is an observed fact about realized losses, not a statement about latent vulnerabilities or tail risk. Profitability is similarly healthy: US banks posted 2025 ROE of 11.39% and ROA of 1.17% (FDIC) (Observed, A); euro-area banks 8.96% and 0.68% (ECB CBD2) (Observed, A); UK banks 9.20% and 0.68% (IMF FSI) (Observed, B), with the 2008 UK ROE reading excluded from the comparison as inconsistent with contemporaneous evidence, a disclosed exclusion.

5.2 Delinquencies and credit conditions

Bank-book delinquencies sit far below crisis peaks. US bank CRE delinquency (excluding farmland) was 1.50% in Q2 2026, versus the GFC peak of 9.13% in Q1 2010 (Observed, A); bank-book credit-card delinquency was 2.85% (seasonally adjusted), down from the 2024 peak near 3.2%; all-loan delinquency was 1.38%. The NY Fed's Q2 2026 household data (all lenders) show flow into serious delinquency at 6.97% for credit cards, 3.00% for auto loans, and 1.52% for mortgages, with total household debt of $18.771tn (Observed, A). One desk calculation is flagged as provisional: auto-loan balances 90+ days delinquent at 5.49%, the second-highest of 94 quarterly observations since 2003 and above the GFC-era peak of 5.27% in Q4 2010 (Reconstructed, B), pending verification in the NY Fed workbook.

The CRE stress that bank books have not yet recognized sits in the CMBS market. Overall CMBS special servicing reached 11.42% in August 2026, the highest since February 2013 (Observed, A) (Trepp); office special servicing was 16.90%, with $3.16bn across 32 loans transferred in August alone. Office CMBS delinquency is near 12% and multifamily 7.69% (trade-press reporting of Trepp/Bloomberg data (Observed, B)); office vacancy was 18.3% nationally in Q2 2026 (CBRE) with 12.6m sq ft of net absorption and a 30bp quarterly vacancy decline. The maturity wall is the forcing mechanism: about $5.0tn of CRE debt is outstanding, with $875bn (17%) maturing in 2026 and $652bn in 2027 (MBA survey (Observed, A)); of the 2026 maturities, depositories



Figure 8. US aggregate PPNR coverage of net charge-offs, 2007–2026. H1 2026 coverage of 6.4x is 4.8 times the 2009 trough of 1.33x. This measures realized-loss absorption, not latent risk.

Source: FDIC Summary API (Observed, A). Note: 2026 is H1 annualized; the 2009 net-income series uses the FDIC's restated figures.

hold $396bn, CMBS/CLO/ABS $200bn, credit companies and warehouse lines $163bn, life insurers $76bn, and agency multifamily $39bn. Depositories are the largest single holder of the 2026 wall.

Lending standards have normalized. The SLOOS net percentage of banks tightening C&I standards for large and middle-market firms was 0.0% in July 2026, down from 50.8% in Q3 2023 and the series peak of 83.6% in Q4 2008 (Observed, A). The 2025 Shared National Credit review (released January 2026) shows 6,857 borrowers and $6.9tn of commitments (+6% year over year) with a non-pass share of 8.6% versus 9.1% in 2024, though the agencies state the decline came mainly from new-commitment growth, not credit-quality improvement (Observed, A). US banks hold 45% of commitments but only 22% of non-pass loans; nearly half of commitments are leveraged and leveraged loans are 81% of non-pass loans, so the weakest credits sit disproportionately outside the banking system.

5.3 Bank linkages to non-bank financial intermediaries

Bank loans to NDFIs reached $2.026tn in August 2026 for all commercial banks, up 19% year over year from $1.700tn (Observed, A) (Fed H.8), and $2.047tn in the week ending 16 September; the Q2 2026 year-over-year rate was +22.4%. An April 2026 reporting clarification reclassified several H.8 categories, so growth spanning that break mixes definitional and real change and carries a comparability warning. Less-traditional lending at the eight US GSIBs was 12.6% of assets in June 2026, up from 10.1% a year earlier and 5.4% in June 2011 (St. Louis Fed (Observed, B)). No official aggregate exists for subscription lines, NAV loans, or CLO and private-credit warehouse facilities; this is a measurement gap, and the paper does not infer hidden leverage from its absence.

The growth of the measured NDFI linkage is the channel through which a private-credit repricing would reach bank balance sheets (Section 9.3).

6 Digital runs and the 2023 autopsies

The 2023 failures are used here as calibrated precedent for what a run costs in time and dollars, verified against official records. They are bucket (a) history, and their mechanism, solvency concern leading to liquidity flight leading to forced loss realization, is the template for the paper's scenarios and contagion paths.

Silicon Valley Bank: on March 8, 2023 the bank announced a $1.8bn loss on securities sales plus a planned capital raise; on March 9 depositors withdrew $42bn in a single day and the bank ended the day with a negative cash balance of $958m (California DFPI regulatory notice); it was closed on March 10. About 94% of deposits were uninsured. The belated balance-sheet repair announcement triggered the run it was meant to prevent; social media and venture-capital channels coordinated withdrawals in hours (Barr, 2023).

Signature Bank: on Friday, March 10, 2023 customers withdrew $18.6bn, 20% of total deposits (FDIC OIG material loss review); expected Monday March 13 outgoing wires grew to $7.9bn, and the bank could not pledge sufficient preapproved collateral at the discount window to cover a $3.4bn end-of-day settlement shortfall, despite $23.5bn of pledged borrowing capacity at the New York Fed and FHLB plus $4.6bn of repo lines, because collateral eligibility and diligence requirements could not be satisfied in time (Federal Deposit Insurance Corporation, 2023). About 76% of deposits were uninsured; between March 7 and March 17 the bank lost more than half its deposits. The FDIC also reported that about 40% of its New York-region large-bank supervisory positions had been vacant or temporary since 2020.

First Republic Bank: Q1 2023 deposit outflows exceeded $100bn, roughly 40–50% of deposits, with realized net outflows of about $103.6bn from March 10 to April 28, exceeding the bank's stress-test estimates across every scenario; its stress tests overestimated uninsured-deposit stickiness. On March 16, eleven of the largest banks placed a $30bn consortium deposit; excluding the infusion, the deposit loss would have been nearly 54%. About 74% of deposits were uninsured. The bank failed on May 1, 2023 with about $229bn in assets, the second-largest US failure; JPMorgan Chase assumed the deposits and most assets at an estimated $13bn cost to the Deposit Insurance Fund.

Credit Suisse: net asset outflows of CHF 110.5bn in Q4 2022 and CHF 61.2bn in Q1 2023 (bank earnings release via SEC EDGAR); on March 19, 2023 FINMA ordered the write-down of about CHF 16bn of AT1 capital notes, lifting the CET1 ratio from 14.1% to 20.3%, and the emergency merger with UBS was announced the same day. The mechanism was years of scandals and losses destroying franchise confidence: wealth-management client asset flight, not a classic deposit run, with the March 2023 global banking stress as accelerant rather than cause.

The FDIC staff study of the 2023 runs found uninsured balances of about 94% (SVB), 76% (Signature), and 74% (First Republic); the top 0.5% of depositors controlled 62% of deposits at Signature, 50% at First Republic, and 39% at SVB; concentrated uninsured depositors evacuated fastest; the vast majority of outflows left by wire; and daily outflows exceeded 20% of deposits, which the FDIC calls unprecedented. Cookson et al. (2023) document the coordination technology. Two structural facts carry forward to 2026. First, the mechanism is unchanged: solvency concern, liquidity flight, forced loss realization, in that order. Second, discount-window readiness remains the

untested link: both 2023 failures involved insufficient pre-positioned collateral, interagency guidance now encourages readiness, but stigma and operational readiness at 2023 speeds are unproven.

Reforms implemented since: the systemic risk exception for SVB and Signature (March 12, 2023, all depositors made whole); the FDIC special assessment, final rule November 2023, at 13.4 basis points for eight quarterly periods, initially sized at $16.3bn and later adjusted to $20.4bn; and 2023 interagency statements on discount-window preparedness. Proposed but not implemented, with status to be verified before any policy use of this paper: the long-term debt requirement for large banks (proposed August 2023), broader AOCI inclusion in regulatory capital (Basel III endgame proposal), and deposit insurance reform (FDIC May 2023 options report, no legislative action).

7 Historical framing

Three comparisons discipline the reading of current conditions: earnings buffers against 2008–2009, the current price pattern against past stress episodes, and what is new in 2026.

First, earnings buffers. Aggregate PPNR covered net charge-offs 6.4 times in H1 2026, versus 1.33 times at the 2009 trough, 1.46 times in 2010, and 2.01 times in 2008 (Observed, A). The current buffer is 4.8 times the GFC nadir. Post-crisis norms ran 6–7 times from 2014 to 2020; the Covid-era spike to 11.6–12.3 times in 2021–2022 was an artifact of collapsed charge-offs. The earnings buffer is far stronger than at the GFC nadir; this is an observed fact about realized losses.

Second, the price pattern. The current configuration, price stress in duration and bank equity valuations with funding plumbing functioning normally, is the opposite of 2008 and 2020, when funding spreads led, and closer to 2022–2023, when rate moves led and funding followed only in the March 2023 episode. Money-market gauges (SOFR tail 9bp, EURIBOR-DFR 26bp) confirm the plumbing reading.

Third, what is new. Digital-run speed is the structural change: hours not days, 24/7, social-media coordinated, with 2023 daily outflow rates unprecedented in FDIC records. Private-credit opacity is the measurement change: bank NDFI lending and GSIB less-traditional exposures keep growing while warehouse and subscription-line facilities have no official aggregate. AI-capex concentration is the correlation change: large technology borrowers and datacenter exposures tie correlated credit risk to a single investment cycle, the subject of KR-WP-2026-10 (Kilambi Research, 2026a) and the contagion path in Section 9.3. What rhymes with earlier episodes: funding fragility through confidence-sensitive deposits, correlated assets (then subprime MBS, now office CRE and underwater securities), and the euro-area sovereign-bank nexus, still present in 2026.

8 Stress scenarios

Everything in this section is bucket (c): hypothetical tails, not predictions and not observations. Six scenarios were run through 26 banks: all G-SIBs present in the three reconstructed datasets (23), plus the three non-G-SIBs with the most acute named latent vulnerabilities, PNC (narrowest CET1 headroom among large US regionals at 9.9%), Truist (largest disclosed FHLB reliance at $17.6bn advances), and TD (46% variable-rate Canadian mortgage book). Each row of the scenario table (Appendix A.4) records the first pressure point, the mechanism, a reliance- or headroom-ranked severity (Low, Medium, High, or Unknown, which is a severity of undisclosed inputs, not a statement

of safety), the computed metric with its evidence grade, and the explicit missing input. The scenarios are run in mechanism order, because the binding sequence is the finding: funding first, capital second.

S1: rate shock, +200bp. Pressure emerges first in unrealized securities losses, not capital. The observable baseline is the current drag: (HTM unrecognized loss + |AOCI|) / TCE ranges from 4.6% (M&T) to 40.6% (Bank of America, $82.1bn HTM loss) (OBSERVED, A/C). The +200bp increment itself is not estimated: portfolio duration and hedge coverage are undisclosed for every US bank (Section 4.1 gap). The mechanism that converts a rate shock into stress is funding-side, not accounting-side: wider unrealized losses raise wholesale counterparty haircuts and rating attention, and any deposit outflow that forces sales crystallizes the marks into TCE, which is the link to S2. Highest severity by baseline drag: Bank of America, Wells Fargo, Citi, U.S. Bancorp, Truist. TD is High by a different mechanism: its 46% variable-rate book reprices borrower payments immediately. For Mizuho the disclosed book is −387.0bn JPY on foreign bonds (June 2026), offset by +2,359.6bn JPY in Japanese-stock gains; MUFG and SMFG disclosed no June 2026 breakdown, so both are Unknown. For EU banks the increment is not quantified: the last supervisory aggregate is the ECB's February 2023 figure.

S2: 15% uninsured-deposit outflow. The 15% pace is survivable wherever LCR exceeds 100%, since LCR calibration already assumes 30-day stress runoff faster than this. The binding mechanism is loss crystallization: outflows beyond AFS sales force HTM sales, turning S1 marks into TCE losses. That is the documented 2023 mechanism (Section 6). Dollar outflows at 15% run to about $261bn of uninsured deposits at JPMorgan (RECONSTRUCTED, C). Severity is High only above 60% uninsured (JPMorgan, Citi on the FDIC-proxy basis). The custodians show the highest uninsured shares by construction; their deposits are institutional operating balances, stickier than the share implies, but 2023 showed institutional flight reverses that stickiness fast. Missing for every bank: HQLA level and composition and the 30-day contractual wholesale maturity ladder, so no dollar liquidity gap is computed for any bank.

S3: credit losses at 2x current allowances; S5: recession (unemployment +3pp, GDP −2%). Bank-level allowances are disclosed for only three US banks, so 2x-allowance arithmetic cannot be run directly; the calibrated proxy is the supervisory scenario-minimum CET1 (Fed 2026 severely adverse for US banks, EBA 2025 adverse for EU banks), which embeds severe credit losses (MODELLED, A) (Board of Governors of the Federal Reserve System, 2026; European Banking Authority, 2025). No bank breaches the 4.5% minimum in the calibrated tail. Narrowest headroom: Societe Generale 8.75% (425bp over minimum), BNP Paribas 9.49%, Wells Fargo 9.20%, Truist 9.70%, U.S. Bancorp 9.80%, Bank of America 9.90%. Two caveats: the EBA test has no predefined pass-fail threshold and MDA headroom was not verified against bank-specific thresholds, so "no breach" is against the 4.5% minimum only. HSBC, Barclays, UBS, and all Japanese and Chinese banks have no comparable adverse tail: Unknown. Chinese banks report 203–290% allowance coverage of reported NPLs, but the LGFV loss surface is undisclosed by every sample bank; the tail is losses exceeding disclosed coverage by multiples, and that caveat applies to every China row. In the recession scenario, first pressure is credit migration: Stage 2 builds and allowance charges hit earnings before capital. Current PPNR covers net charge-offs 6.4x versus 1.33x at the 2009 trough, so the earnings buffer is 4.8x the GFC nadir. The banks where recession converts a latent vulnerability into losses fastest are TD and CIBC (variable-rate and renewal-wave mortgages; arrears 41–51bp and rising) and the Australian four (arrears 1.01%, interest-only origination 23.5%, the DTI cap the binding backstop).

S4: office CRE impairment, −30%. At the stated assumption (−30% values, 50% loss-given-

default, hence a 15% loss rate on disclosed office exposure), this is an earnings event, not a capital event, everywhere it can be quantified (MODELLED, B): JPMorgan $14.7bn office exposure implies about $2.2bn of loss, 0.69% of TCE; Bank of America $11.4bn implies about $1.7bn, against $2.8bn already criticized (24.6%) and $58m of H1 net charge-offs, meaning part of the tail is already reserved and shrinking. Ten of sixteen US banks disclose no office figure, and no EU/UK/CA/APAC bank in the set discloses office exposure except through the Hong Kong CRE pocket: Unknown for 20 of 26 banks, where absence of disclosure is not evidence of zero exposure.

S6: wholesale funding freeze, 90 days. LCR covers 30 days, not 90; days 31–90 must roll into FHLB advances or the discount window. No bank discloses a 90-day contractual maturity ladder, so no dollar liquidity gap is computed and severity is reliance-ranked. Truist is High ($17.6bn FHLB advances outstanding); Huntington ($27.6bn wholesale funding, +$3.2bn), Citizens ($11.6bn of $30.5bn FHLB capacity), and M&T ($4.6bn) are Medium. The 2023 lesson stands: banks chose FHLB and the BTFP over the discount window (stigma). A freeze that also shuts FHLB would be the unprecedented leg. For Japanese megabanks the dollar-funding leg is latent (SMFG holds JPY 9.2tn of foreign debt securities; the split is undisclosed). In Europe, TLTRO is fully repaid, funding is market-based with LCRs of 125–161%, and first pressure is euro commercial-paper rollover spreads.

Figure 9 summarizes first pressure points across banks and scenarios; Table 2 collects the binding constraints the scenarios identify, in order.

Table 2. Binding constraints identified by the scenario exercise, in order

Rank	Constraint
1	HTM loss crystallization under forced sale (S1+S2 interaction): unrealized marks become TCE losses only when outflows force sales.
2	Uninsured-deposit concentration at money centers and custodians (S2): the 2023 mechanism, at 2026 shares.
3	Wholesale and FHLB reliance at Truist, Huntington, and Citizens (S6): days 31–90 of a funding freeze have no disclosed maturity ladder.
4	Narrow calibrated-tail headroom at Societe Generale (8.75%), BNP Paribas (9.49%), and Wells Fargo (9.20%) against the 4.5% CET1 minimum (S3/S5).

Notes: severities are reliance- or headroom-ranked, not probabilities. No scenario produces a capital shortfall against the 4.5% CET1 minimum where the scenario can be calibrated. Liquidity gaps in dollars cannot be computed for S2/S6. Source: desk scenario engine (MODELLED, B).

9 Contagion and backstops

Contagion paths are bucket (c), hypothetical. Sixteen hops across three paths are tabulated with mechanisms and evidence grades in `data/contagion/contagion_paths.csv`; the text below states each path's logic and its terminal point. The amplifier common to all three paths is a rate shock hitting US HTM books, Japanese megabank foreign-bond books, and EU amortised-cost books at

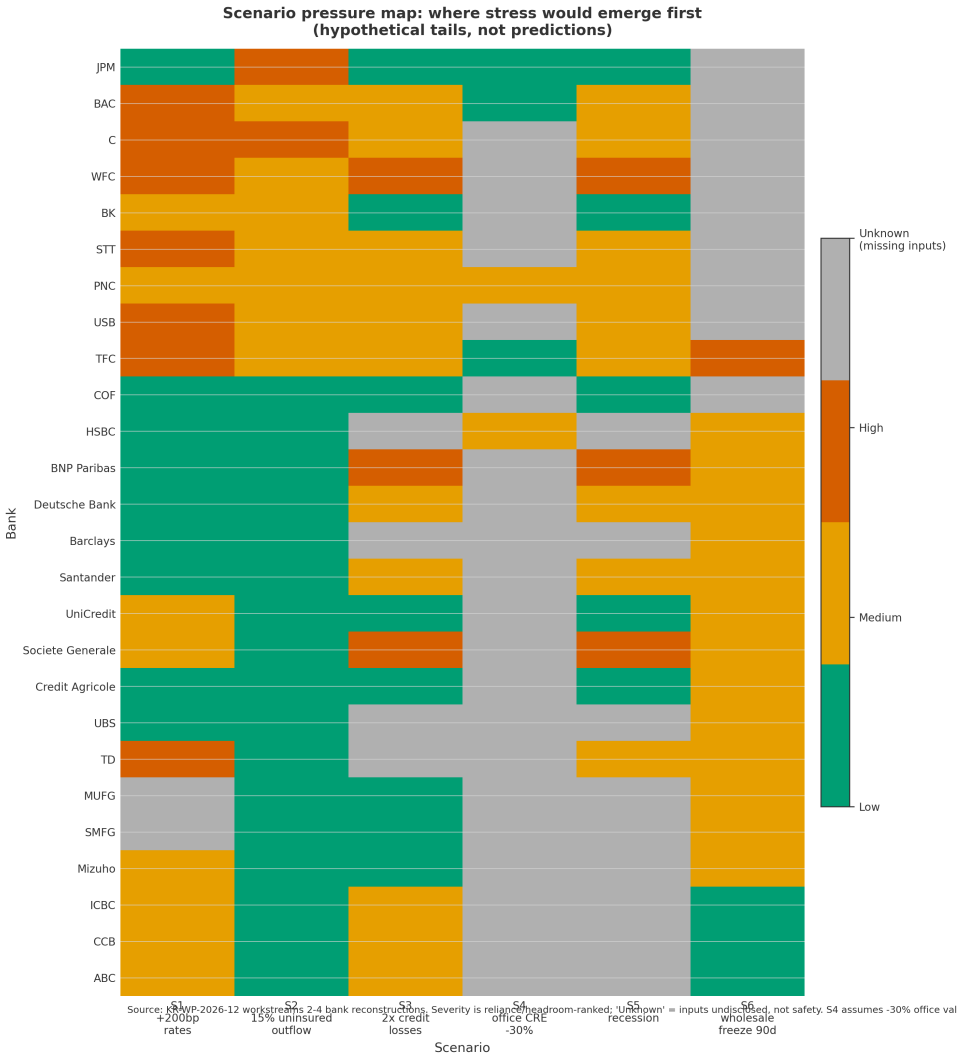


Figure 9. First pressure point by bank and scenario, 26 banks. Severity is reliance- or headroom-ranked, not a probability; Unknown means inputs are undisclosed, not that the bank is safe.

Source: desk scenario engine, `data/scenarios/scenario_results.csv` (MODELLED, B). Note: scenario inputs are stated assumptions (rate shock +200bp, 15% uninsured outflow, 2x allowances, office −30%, recession, 90-day wholesale freeze). Supervisory tails are used as the S3/S5 calibration.

the same time, through shared wholesale markets: FHLB in the US, euro commercial paper, and dollar funding of Japanese banks.

Path A: US regional liquidity to HTM crystallization (the 2023 rhyme). A rate shock widens unrealized HTM losses (baseline: Bank of America $82.1bn); the marks become salient and uninsured corporate depositors reassess, with the observed 2023 precedent of SVB losing $42bn in one day coordinated over messaging apps; outflows beyond AFS sales force HTM sales and the marks crystallize into TCE; TCE depletion draws negative rating action while FHLB and wholesale counterparties raise haircuts and spreads, with reliance concentrated at Truist ($17.6bn FHLB advances) and Huntington ($27.6bn wholesale funding); peer regionals catch it three ways, through the same duration risk on their own books (2026 marks undisclosed), the shared FHLB backstop (simultaneous draws against pledged collateral), and depositor flight to money centers and money funds (the 2023 pattern). Terminal: FDIC purchase-and-assumption, as with First Republic to JPMorgan in 2023.

Path B: euro-area sovereign-bank nexus. Italian or Spanish spread widening hits domestic sovereign books held at amortised cost (BBVA 6.84%, UniCredit 5.21%, Intesa Sanpaolo 4.85% of exposure, June 2025 lower bound); OCI depletion plus wholesale spread widening for Italian and Spanish banks reprices euro commercial-paper rollover, with adverse-tail CET1 of 8.75% (Societe Generale) and 9.49% (BNP Paribas); cross-border interbank retrenchment from periphery names follows. The standing backstop is ECB standard operations (MRO, LTRO) with broad collateral including credit claims, plus ELA via national central banks (tested: Greece 2015); TLTRO III matured in December 2024 with no targeted successor. Terminal tail: SRM resolution, tested on Banco Popular (2017) and Sberbank Europe (2022), but bail-in of senior unsecured at G-SIB scale is untested in the EU. Credit Suisse was resolved under Swiss emergency law, not the SRM.

Path C: AI and datacenter credit repricing via private credit. This path is cross-referenced, not duplicated: KR-WP-2026-10 (Kilambi Research, 2026a) owns the buildout credit analysis, and no exposure data was rebuilt here. The first observation of AI project-risk repricing is already dated: Meta Hyperion JV senior secured notes ($27bn, 80% Blue Owl-owned project company) traded at 94.4 cents in September 2026 (Observed, B) (KR-WP-2026-10). Private credit funds ($2.5tn global, BIS; US technology direct lending above $1tn, 44% of direct lending, per KR-WP-2026-10) face marks and redemptions; banks' indirect exposure runs through revolvers and subscription and warehouse lines to NDFIs, with US bank lending to NDFIs up 22.4% year over year in Q2 2026. State Street originated $1.14bn of CLO loans in Q2 2026 and holds $0.97bn of non-US CLO securities. If private credit pulls back, borrowers draw bank revolvers, so the liquidity hit and the credit hit arrive together; correlated borrowers include AirTrunk's JPY 191.6bn Tokyo loan (12 banks) and $2.325bn Johor facility (30 lenders), and Princeton Digital Group's $750m sustainability-linked facility. KR-WP-2026-10 finds no aggregate crowding out to date (US credit expanded, spreads narrowed); the distributional tail is rationing of smaller bank-dependent borrowers. Roughly $1.65tn of off-balance-sheet AI obligations at the five largest names (press-reported (Assumed, C)) sits outside standard aggregates: part of the loss surface is unmeasured.

9.1 Backstops: tested versus assumed

The backstop inventory is stated plainly as tested or assumed. Proven in 2023: FHLB surge capacity; BTFP par-value lending (9,000 loans to 1,327 borrowers, per Bank Policy Institute analysis of the released data); FDIC purchase-and-assumption with the systemic risk exception; and SNB emergency liquidity to Credit Suisse (with the CHF 16bn AT1 writedown under Swiss emergency

law, outside the EU resolution framework). Gone since: the BTFP ceased new loans on 11 March 2024 and the last loan was repaid on 11 March 2025 (Federal Reserve); the par-value facility that separated liquidity stress from HTM loss crystallization no longer exists. ECB TLTRO III matured in December 2024 with no successor.

Assumed but untested: discount-window borrowing without stigma (the 2023 revealed preference ran the other way); single-point-of-entry execution under orderly liquidation authority (no US G-SIB has failed; 2023 used ad-hoc purchase-and-assumption, not SPOE); long-term debt rules for large regionals (proposed, not final); ECB standard operations absorbing a systemic bank run post-TLTRO; the Bank of England's Contingent Term Repo Facility (never activated); and bail-in of senior unsecured at G-SIB scale in the EU. Central-bank swap lines are proven (2008, 2020, 2023) but priced as penalty facilities.

The bottom line on backstops has four parts. First, the instrument that made 2023 survivable for HTM-heavy banks is closed: a repeat of the solvency-concern to liquidity-flight to forced-sale sequence now runs through the discount window (stigma) and FHLB (collateral-constrained), not through par funding. Second, US regional resolution at scale still routes through ad-hoc purchase-and-assumption; SPOE is doctrine, not practice. Third, the euro area enters any stress with TLTRO gone and no targeted replacement; the ECB's crisis liquidity toolkit is back to standard operations plus ELA, neither tested at systemic scale since the 2010s. Fourth, cross-border dollar funding of Japanese megabanks and euro commercial paper have no standing cross-currency backstop beyond the swap lines.

10 Monitoring dashboard

The desk maintains a 13-indicator dashboard (Table 3, Figure 10), rebuilt from workstreams 1–5 datasets with no new external calls. Each indicator has a current reading, a historical stressed level, and a warning threshold that is a stated judgment call, not a model output; the chart places the current reading on a gauge from a calm anchor (left) to the historical stressed level (right), with the black tick as the warning threshold. Bars are gauges, not forecasts. Lead indicators (market prices, repricing in days): US 10Y 90-day change, KBE drawdown, bank CDS (not retrieved: a stated gap), AT1 new-issue coupon. Coincident indicators (balance-sheet quantities that bind during stress): HTM/TCE, uninsured-deposit share, NDFI lending growth, FHLB reliance. Lagging indicators (confirming stress after it arrives): LCR minimum, PPNR coverage, NPL ratios, mortgage arrears, office criticized share.

The verdict: no indicator is at its historical stressed level. Two sit at or past their warning thresholds: the minimum disclosed LCR (107% versus 110% warn, State Street) and the maximum uninsured-deposit share (82.5% versus 70% warn, on a grade-D FDIC proxy for Citi). Both are structural features of the banks involved (custody model; FDIC-proxy measurement) rather than new deterioration, but they mark where a confidence event would bind first. Two more are within one move of warning: the US 10Y 90-day change (+80bp versus +100bp warn) and the KBE drawdown (−10.1% versus −15% warn). The earnings and asset-quality gauges (PPNR 6.4x, EBA NPL 1.8%, Canadian arrears 0.51%) read calm. The dashboard's blind spot is the CDS leg, which was not retrieved.

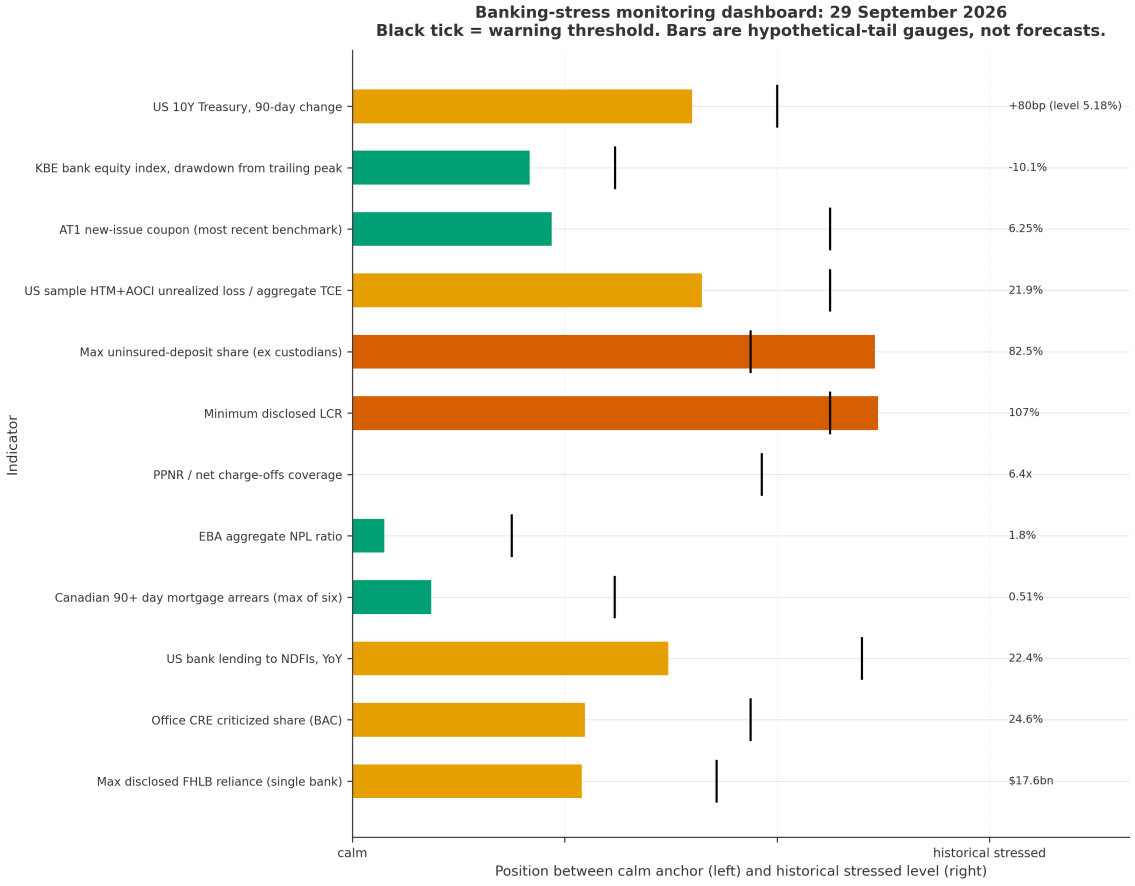


Figure 10. Banking-stress monitoring dashboard: current readings against warning thresholds (black ticks) on gauges from calm (left) to historical stressed levels (right). No indicator is at its historical stressed level; two sit at or past warning; two are within one move.

Source: workstreams 1–5 datasets. Note: thresholds are judgment calls; bars are gauges, not forecasts. The CDS leg was not retrieved (stated gap).

Table 3. Banking-stress monitoring dashboard, 29 September 2026

Indicator	Current reading	Warning	Stressed anchor
US 10Y, 90-day change	+80bp (level 5.18%)	+100bp	–
KBE drawdown from trailing peak	−10.1%	−15%	−36.4% (Mar 2023)
US IG bank 5Y CDS	not retrieved	150bp	–
AT1 new-issue coupon	6.25% (UniCredit, 18 Sep)	8% or pulled	market shut (Mar 2023)
US HTM+AOCI loss / aggregate TCE	21.9% (sample max 40.6%)	30%	40% (judgment)
Max uninsured-deposit share, ex custodians	82.5% (Citi, proxy)	70%	94% (SVB)
Minimum disclosed LCR	107% (State Street)	110%	100% (minimum)
PPNR / net charge-off coverage	6.4x	3.0x	1.33x (2009)
EBA aggregate NPL ratio	1.8%	3.0%	7.5% (2013)
Canadian 90+ day mortgage arrears	0.51% (CIBC, max)	1.0% at >1 bank	–
US bank lending to NDFIs, YoY	+22.4% (Q2 2026)	30%	–
Office CRE criticized share	24.6% (BAC)	35%	–
Max disclosed FHLB reliance	$17.6bn (Truist)	$25bn	–

Notes: thresholds are judgment calls; see `data/dashboard/dashboard.csv`. Citi's 82.5% is a grade-D FDIC lead-bank proxy. M&T's LCR is about 106% estimated. Source: workstreams 1–5 datasets (FRED H.15, Yahoo Finance, bank filings, EBA, OSFI via filings) (OBSERVED, A/B).

11 Falsifiers

The paper's "no observable system-level stress" conclusion is falsified by any of the following: any sample bank cutting or suspending its announced buyback or dividend; CET1 falling toward MDA thresholds at any sample bank; LCR approaching 100% at any bank; the EBA aggregate NPL ratio doubling from 1.8%; Canadian 90+ day mortgage arrears breaching 100bp at more than one bank; an AT1 new issue pulled or priced at distressed levels; a supervisory disclosure of material unrealized-loss-driven capital action; reactivation of a BTFP-like par facility or invocation of the systemic risk exception; a SOFR tail-gauge spike toward 2019 levels; EURIBOR-DFR spreads widening back toward 2022–2023 peaks; CMBS office delinquency migrating into bank-book charge-offs at scale; or PPNR coverage collapsing toward 3x. The latent-vulnerability inventory is falsified in the other direction by their resolution: a full 2026 supervisory aggregate showing EU unrealized losses fully hedged or immaterial; disclosure of LGFV exposures with coverage intact; or published HQLA and maturity-ladder data showing days 31–90 of a wholesale freeze covered. The hypothetical-tail scenarios are falsified by their own calibration: a realized credit cycle that stays milder than the Fed severely adverse or EBA adverse paths while capital ratios hold would show the tails were conservative, not that the mechanism analysis was wrong. The dashboard in Table 3 is the instrument for tracking these readings as new data arrive.

12 Limitations

Retrospective and non-pre-registered. This paper is retrospective: the datasets were assembled after the conditions described were underway, the hypotheses were formed with knowledge of the outcomes, and no analysis plan was pre-registered. The falsifiers in Section 11 are stated prospectively to discipline future updates, but they do not cure the retrospective design. Readers should discount the sharpness of the results accordingly.

Data vintage. FDIC panel data end at 2026-Q1 while equity and market data run to 29 September

2026; EBA aggregates are Q4 2025; the last supervisory aggregate for EU unrealized securities losses is February 2023; the office-vacancy series is a Q2 2026 snapshot. Primary OSFI and Bank of Canada 2026 mortgage-risk publications were not retrieved; secondary aggregates were quarantined, not used.

Measurement. Citi's 82.5% uninsured-deposit share and several regional-bank ratios are FDIC lead-bank subsidiary proxies (grade D), not consolidated BHC figures. The deposit beta is a cost-of-funds beta: FDIC interest expense includes wholesale expense. The sovereign-concentration ratio uses a vintage-mismatched denominator (December 2024 exposure), making it a lower bound. The NY Fed auto-loan 90+ stock statistic (5.49%) is a desk calculation pending workbook verification (grade B). The UK 2008 ROE reading was excluded from the comparison chart as inconsistent with contemporaneous evidence, a disclosed exclusion. H.8 NDFI lending growth spanning the April 2026 reclassification mixes definitional and real change.

Coverage gaps. No bank discloses a 90-day contractual wholesale maturity ladder or full HQLA composition, so no dollar liquidity gap is computed in S2/S6. Portfolio duration and hedge coverage are undisclosed for every US bank, so the S1 rate-shock increment is not estimated. Single-name bank CDS and current iTraxx Senior Financials were not retrieved. Fed discount-window, standing repo, and ON RRP usage, and ECB emergency facility usage, were not fetched. Cross-currency basis was not fetched. CRE concentration by bank, leveraged-lending retained exposure, and CLO warehousing pipelines are covered only partly by SNC and SLOOS. UK banks and UBS are not in the EBA stress test, so no adverse tail exists for them. Chinese LGFV exposures are undisclosed by every sample bank.

Scope. The paper measures banking-system stress, not the probability of a recession, the path of policy rates, or election outcomes. It does not model deposit-flight dynamics beyond the stated S2 assumption, does not haircut HTM portfolios under hypothetical rate paths, and does not project credit losses beyond published supervisory scenarios. The AI and datacenter credit analysis belongs to KR-WP-2026-10 and is cross-referenced, not repeated.

13 Conclusion

The banking system looks capitalized and liquid today, with identified latent vulnerabilities and a deteriorating price environment. That is the finding, stated without uplift. Capital ratios sit above minima across 56 reconstructed banks; liquidity coverage exceeds 100% where disclosed; the June 2026 Federal Reserve stress test left all 32 banks above minimum requirements; the AT1 market is open; buybacks run at full pace. The price environment has deteriorated: the US 10-year at 5.18% above its October 2023 peak, a September tightening wave across four central banks, US bank equities down 8 to 23% from summer highs, the MOVE index above 100, and weak-tail credit dispersion at its widest since 2022, while money-market gauges stay calm.

The binding sequence in a shock is funding, not capital. A rate shock widens unrealized securities losses first; uninsured-deposit flight then crystallizes the marks; the facility that separated those two steps in 2023 no longer exists. Bank of America is the epicenter of the unrealized-loss drag ($82.1bn, 40.6% of TCE); Societe Generale has the narrowest calibrated-tail headroom (8.75% against the 4.5% minimum); Truist carries the largest disclosed FHLB reliance ($17.6bn). No calibrated tail breaches the minimum at any of the 26 scenario banks. The dashboard's thirteen indicators show none at historical stressed levels, two at warning on structural features rather than new deterioration, and two within one move. The falsifiers in Section 11 say what would change this verdict; the

limitations in Section 12 say what the data cannot show. The null is reported plainly because the data support it, not because the risks are absent.

Acknowledgments

Production note. This paper was produced on 2026-09-29 by Kilambi Research's quantitative research pipeline with AI assistance; all analysis was verified against source data, and every number in the text traces to a dataset row or a stated model input. No part of the analysis was pre-registered. Workstream research notes, datasets, and reproduction scripts are archived under `~/workspace/ai-banking-stress/` as listed in the data appendix.

A Per-bank and scenario tables

Table 4. US banks, Q2 2026: headline ratios

Bank	CET1	Stress min	Uninsured	LCR	HTM loss
JPMorgan	14.2%	12.6%	64.3%[c]	110%	$18.2bn
Bank of America	11.2%	9.9%	45.4%[d]	112%	$82.1bn
Citigroup	12.8%	10.3%	82.5%[d]	113%	n/d
Wells Fargo	10.26%	9.2%	50.6%[d]	n/d	$32.5bn
PNC	9.9%	10.3%	45%	n/d	n/d
U.S. Bancorp	10.8%	9.8%	53.1%[d]	n/d	$9.5bn
Truist	10.9%	9.7%	44.9%[d]	113%	$8.2bn
Capital One	13.7%	11.0%	15%	165%	n/d
Citizens	10.4%	8.6%	47.5%[d]	n/d	n/d
Fifth Third	9.93%	9.7%	40.2%[c]	n/d	n/d
M&T	10.19%	9.2%	46.5%[c]	∼106%[e]	n/d
KeyCorp	11.2%	9.9%	45%[c]	n/d	n/d
Regions	10.72%	10.3%	39.2%[d]	n/d	n/d
Huntington	10.0%	9.3%	∼31%	n/d	n/d
BNY Mellon	11.0%	11.8%	99.3%[d]	111%	n/d
State Street	10.8%	10.8%	n/d[f]	107%	n/d

Notes: CET1 is the BHC standardized-approach ratio (OBSERVED, A). Stress-test minima are the Federal Reserve 2026 severely adverse scenario (MODELLED, A). Uninsured shares: (c) computed from filings, (d) FDIC lead-bank proxy, otherwise company-stated. (e) estimated. (f) FDIC ratio exceeds 100% on a denominator artifact; no consolidated ratio published. n/d: not disclosed in the Q2 2026 filing. Source: `data/bank_us.csv`.

B Data appendix

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Table 5. Europe, UK, and Canada: headline ratios

Bank	CET1	Requirement	LCR	Notable
HSBC	14.8%[g]	–	142%[g]	$1bn buyback reinstated
BNP Paribas	13.0%	–	–	adverse CET1 9.49%
Deutsche Bank	14.2%	–	139%	AT1 announced 21 Sep 2026; adverse 10.23%
Barclays	14.0%	–	161%	£1bn buyback
Santander	12.7%	–	152%[g]	€1.8bn buyback approval; adverse 10.91%
UniCredit	15.2%	–	–	€750m AT1 at 6.25%, 18 Sep 2026
Societe Generale	13.1%	–	–	adverse CET1 8.75%; €1.5bn buyback
Credit Agricole	17.2% (Group)	–	–	adverse 11.01%
ING	13.9%	–	–	adverse 10.63%
UBS	14.3%	–	–	$3bn buyback through 2Q27
Intesa Sanpaolo	13.1%	9.96%	–	€9.4bn 2026 shareholder return guide
BBVA	12.90%	8.98%	–	adverse 11.02%
NatWest	13.6%[g]	–	–	–
Lloyds	13.7%[g]	–	–	£1bn buyback
Nordea	15.4%	13.8%	–	adverse 12.62%
Danske Bank	18.7%[g]	–	–	adverse 13.56%
RBC	13.2%[g]	–	125%	C$4bn Q3 return; arrears 41bp
TD	13.5%[g]	–	–	46% variable RESL; C$13bn FY27 buybacks
BMO	13.1%[g]	–	–	–
Scotiabank	13.3%[g]	–	–	C$8.3bn 12m return; cost of risk 56bp
CIBC	13.0%[g]	–	–	arrears 51bp; C$1.366bn net impaired mortgages
National Bank of Canada	13.6%[g]	–	–	–

Notes: Q2 2026 (Europe/UK), Q3 FY2026 (Canada) (Observed, A/B). (g) rounded from filing or press-mirror values; exact figures in `data/bank_europe_canada.csv`. Adverse CET1 is the EBA 2025 adverse scenario (Modelled, A); UK banks and UBS are not in the EBA test. BMO mortgage rows are grade C (unofficial mirror).

Table 6. Asia-Pacific: headline ratios

Bank	CET1	NPL	NIM	Notable
MUFG	12.95%	0.96%	–	buffer ∼50bp (Shriram stake)
SMFG	12.63%	–	–	JPY 9.2tn foreign debt securities
Mizuho	9.9%[h]	0.80%	–	foreign bonds −387bn JPY; net +1,823bn JPY
ICBC	13.89%	1.31%	1.29% (H1 26)	NIM −83bp since 2021; coverage 213.6%
CCB	14.34%	1.33%	1.37% (H1 26)	LCR 138%; coverage 239.4%
ABC	–	1.25%	1.28% (H1 26)	CAR 17.50%; CNY160bn placement; coverage 290.1%
BOC	–	–	1.27% (H1 26)	profit +5.1%
CMB	–	1.16% (retail)	–	corporate RE NPL 4.47%
CITIC Bank	9.4%	∼1.15%	–	property CNY353bn, −7% H1
Hang Seng	∼16%	–	–	HK$25bn impaired HK CRE
BOCHK	23.80%	0.89% (impaired)	1.48% (H1 26)	profit +7.1%
DBS	16.6%	1.0%	1.87% (Q2 26)	LCR 142; ROE 17.5%
OCBC	15.7%[i]	0.9%	1.70% (Q2 26)	coverage 163%
UOB	15.4%	1.6%	1.74% (Q2 26)	S$615m CRE provisions
CBA	11.6%	–	–	home loans A$637.5bn
Westpac	12.4%	–	–	–
NAB	11.65%	–	1.81% (H1 26)	A$300m provisions
ANZ	12.39%	–	1.54% (Q3 26)	A$175m provisions

Notes: mostly H1 2026 (Observed, A/B). (h) ex unrealized gains/losses. (i) transitional; 14.0% fully phased. n/d rows are disclosure gaps, not signals. China rows carry the LGFV disclosure caveat (Section 4.4). Source: `data/bank_apac.csv`.

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Table 7. Scenario exercise: selection and first pressure points (condensed)

Bank	Selection basis	First pressure point (highest-severity scenario)
JPMorgan	G-SIB	S2: 15% outflow ≈ $261bn uninsured; survivable at LCR 110%
Bank of America	G-SIB	S1+S2: $82.1bn HTM loss (40.6% TCE drag) crystallizes under forced sale
Citigroup	G-SIB	S2: 82.5% uninsured (proxy); S1 drag elevated
Wells Fargo	G-SIB	S3/S5: adverse CET1 9.20%; S1 HTM drag $32.5bn
PNC	narrowest regional CET1	S3/S5: CET1 9.9%; adverse 10.3%
Truist	FHLB reliance	S6: $17.6bn FHLB advances; 90-day freeze is High severity
TD	variable mortgages	S1: 46% variable book reprices; S5: renewal-wave credit migration
Societe Generale	G-SIB	S3/S5: adverse CET1 8.75%, narrowest calibrated headroom
BNP Paribas	G-SIB	S3/S5: adverse CET1 9.49%
Deutsche Bank	G-SIB	S3/S5: adverse CET1 10.23%
Mizuho	G-SIB	S1: −387bn JPY foreign-bond book; offset by stock gains
MUFG / SMFG	G-SIB	Unknown: no June 2026 unrealized-loss breakdown disclosed
Chinese banks	G-SIBs	Unknown tail: LGFV loss surface undisclosed; coverage 203–290% of reported NPLs
HSBC / Barclays / UBS	G-SIBs	Unknown: not in EBA test; no adverse tail

Notes: full 156-row table in `data/scenarios/scenario_results.csv` with mechanism, severity, metric, grade, missing input, and shortfall note per row (MODELLED, B). Severity is reliance- or headroom-ranked, not a probability. "Unknown" means inputs undisclosed. No scenario breaches the 4.5% CET1 minimum where calibrated.

Table 8. Series definitions and reproduction files

Dataset	Contents and reproduction files
Macro series	`data/macro/tidy_*.csv`: 86 ElkassabgiData mirror series; 25 Yahoo tickers; derived policy-rate, yield, funding, and equity files; `macro_summary.csv`
US banks	`data/bank_us.csv`: 249 metric rows, 16 banks, grades A–D; XBRL extracts and 16 archived 10-Q filings in `archive/us_banks/`
Europe/Canada banks	`data/bank_europe_canada.csv`: 278 rows, 22 banks plus reference rows, grades A–C
APAC banks	`data/bank_apac.csv`: 161 rows, grades A–D
Asset quality	`data/asset_quality/`: 13 CSVs covering delinquencies, SLOOS, CMBS, maturity wall, SNC, NDFI, GSIB lending
Profitability	`data/profitability/`: 4 CSVs covering US/EU/UK ROE-ROA and PPNR-NCO
Scenarios	`data/scenarios/`: `bank_selection.csv`, `scenario_results.csv` (156 rows), manifest with input hashes
Contagion	`data/contagion/contagion_paths.csv`: 16 hops across 3 paths with mechanisms and grades
Dashboard	`data/dashboard/dashboard.csv`: 13 indicators with readings, stressed levels, warning thresholds

Notes: raw pulls archived under `archive/` with source and pull date. Build scripts under `scripts/` rerun end to end. Figures in `figs/` were built with the shared institutional chart style (`chart_style.apply()`, `qa_check` passing, 300 DPI, inspected at final size). Every CSV row carries source, evidence grade, and epistemic bucket.

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